Prodigy Anesthesia Quick Review

Cardiomyopathy

What are the chief characteristics of hypertrophic cardiomyopathy?	Patients with hypertrophic cardiomyopathy typically suffer from diastolic dysfunction which is exhibited by increased left ventricular end-diastolic pressures despite an often hyperdynamic left ventricular function. The diastolic stiffness is due to the increased muscle mass of the left ventricle which is typically concentrated in the upper septum just below the aortic valve. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 405.*
What are the symptoms of hypertrophic cardiomyopathy?	Many patients are asymptomatic, but those that do exhibit symptoms exhibit dyspnea on exertion, fatigue, syncope, or angina. The first manifestation of hypertrophic cardiomyopathy in patients younger than 30 years is often sudden cardiac death, which is also the most common cause of death. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 405-406.*
What are the anesthetic management goals for a patient with hypertrophic cardiomyopathy?	Anesthetic goals include the minimization of sympathetic activation to prevent sudden increases in cardiac contractility, expansion of the intravascular volume to avoid hypovolemia, and minimize decreases in left ventricular afterload. Some degree of myocardial depression is often advantageous in these patients and may be obtained through the use of volatile inhalational agents and/or beta-adrenergic blocking agents. Regional anesthesia may worsen the obstruction by decreasing both preload and afterload. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 405.*
What type of cardiomyopathy is seen in patients with aortic stenosis?	In aortic stenosis, the left ventricle must compensate for increased mechanical impedance to ejection (increased afterload). It does so by increasing the muscle mass of the left ventricle known as concentric cardiomyopathy. The remodeling of the ventricular muscle eventually results in decreased compliance of the ventricle and a corresponding decrease in contractility. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 501.*
What type of cardiomyopathy is typically seen in patients with mitral regurgitation?	Eccentric hypertrophy is characteristic of mitral regurgitation. Because of the incompetence of the mitral valve, the left ventricle is exposed to a consistently increased volume of blood. The ventricle compensates by increasing the left ventricular chamber size. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 499.*

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